# DHYANESH MAHENDIRAN

*AI Engineer*

+917806947008 dhyanshm23@gmail.com LeetCode LinkedIn GitHub Coimbatore, India

## PROFESSIONAL SUMMARY

Third-year AI & DS student specializing in full-stack AI development, Generative AI (RAG), and Computer Vision. Experienced in architecting scalable applications using LangChain and FastAPI, and deploying secure, on-premise enterprise solutions with local models (Ollama) and Django. Top 8% in Amazon ML Challenge across 4,000+ teams.

## EDUCATIONS

**Sri Shakthi Institute of Engineering and Technology** | Coimbatore, Tamil Nadu — 2024 - 2028
*B.Tech in AI & DS* — *CGPA : 8.12*

**Kids Club Senior Secondary School** | Tirupur, Tamil Nadu — 2024
*HSC* — *CGPA : 86%*

## PROJECTS

**PDF Readable Knowledge** | Website — Dec 2025 - Mar 2026
*Lead Developer*

- Engineered a **Retrieval-Augmented Generation (RAG)** pipeline to extract, chunk, and semantically query data from unstructured PDF documents.
- Integrated **vector databases** for efficient similarity search, enabling the LLM to generate highly accurate, context-grounded responses with minimal hallucination.
- Implemented a hybrid search strategy combining **BM25 keyword matching** with vector-based semantic search to increase retrieval precision.

***Technologies / Tools Used :*** *Python, RAG, LangChain, Vector Database*

**SLMRouter** | Github — 2026
*Lead Developer*

- **Designed an architecture that treats the model as a deterministic router:** one bounded decision per query, so infinite retry loops are structurally impossible.
- Built a quote-aware parser and strict type coercion that recover **100%** of common small-model formatting errors without an extra model call (**p99 parse time 0.003 ms**).
- Achieved **50/50 correct tool and argument selection** on Llama 3 8B and Mistral 7B (Ollama, llama.cpp).
- Built a deterministic fault simulator (**16 fault classes, 795 cases, zero false executions**) and record/replay of live model runs, enabling CI runs on Python 3.10 to 3.13 without a GPU.

***Technologies / Tools Used :*** *Python, Ollama*

**POSE_FIT** | Website — 2025
*Role: Machine Learning Engineer*

- **Developed a hybrid computer vision fitness application** utilising OpenCV for facial recognition-based user identification and MediaPipe for accurate 3D skeletal posture estimation for SMART INDIA HACKATHON 2025.
- **Formulated biomechanical angle-tracking logic** to dynamically monitor user form (evaluating knee, hip, and torso angles), generating performance scores and automating precise on-screen "REP" counts.
- **Architected a scalable data pipeline**, linking the low-latency Python pose-detection engine to a local MongoDB backend and a Firebase-connected JavaScript/Vite web dashboard for synchronised workout analytics.

***Technologies / Tools Used :*** *Python, OpenCV, MediaPipe, JavaScript, MongoDB, Vite, Firebase*

## SKILLS

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|---|---|
| **Databases :** | PostgreSQL, MongoDB, MySQL, Firebase |
| **Frameworks & Libraries :** | FastAPI, Django, OpenCV, MediaPipe |
| **AI Models & Concepts :** | Llama 3.1, Llama 3.2, RAG, Computer Vision, LLM, SLM |
| **Languages :** | English, Tamil |
| **Programming Languages :** | Python, SQL |
| **Soft Skills :** | Teamwork, Leadership, Critical Thinking, Communication, Adaptability |
| **Tools & Platforms :** | Docker, Git, Vercel, PyMuPDF, Ollama |

## ACHIEVEMENTS AND LEADERSHIP

- **Selected as a Finalist** in the Smart India Hackathon (SIH) for the project POSE_FIT.
- **Showcased MobilityX 2.0 (2025)** [Software Track] — an ML-based EV battery digital twin for health prediction and lifecycle decisions — at the Tamil Nadu Global Startup Summit, Codissia, Coimbatore.
- **Teach Lead of Research & AI Innovation** [2025 - 2028] — Dept. of AI & DS, Sri Shakthi Institute of Engineering and Technology.